

How did the 2017 WHO reclassification of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) affect the risk of malignancy (ROM) estimates across Bethesda thyroid cytology categories in surgical cohorts?

The 2017 WHO reclassification of NIFTP as non-malignant decreased ROM estimates across most Bethesda categories, with the greatest reductions in indeterminate categories III, IV, and V (absolute decreases of 5-42 percentage points), though the magnitude of change varied substantially based on geographic region, with Western institutions showing larger decreases than Asian centers, and institutional factors including surgical selection practices and pathologist interpretation of diagnostic criteria.

Abstract

The 2017 WHO reclassification of NIFTP as a non-malignant entity decreased ROM estimates across most Bethesda cytology categories, though the magnitude of change varied substantially by geographic region, institutional practice patterns, and diagnostic criteria applied. Meta-analytic data from over 25,000 surgical cases demonstrated that NIFTP comprised 4.4% of all resected thyroid nodules, with approximately 79% of cases falling within intermediate cytologic categories (Bethesda III-V). The largest absolute ROM decrease occurred in Bethesda category V (16%; 95% CI: 8-24), while the largest relative decrease was observed in category III (32%; 95% CI: 24-39). Individual institutional studies showed ROM reductions ranging from 2 to 42 percentage points in category V, 7 to 31 percentage points in category IV, and 5 to 20 percentage points in category III, with categories I and VI showing minimal changes.

The impact of NIFTP reclassification was consistently more pronounced in Western than Asian populations, with NIFTP incidence as low as 1.2% in Singapore compared to up to 24.6% of malignancies in some Western series. Inter-institutional variability was largely attributed to differences in surgical selection practices, with a positive correlation between NIFTP rate and institutional resection rate ($r = 0.83$; $P = 0.02$), and pathologist interpretation of nuclear features. Studies applying the stricter 2018 diagnostic criteria found only 2.3% of papillary thyroid carcinomas qualified as NIFTP, with non-significant ROM changes. Despite documented ROM decreases, the clinical impact on surgical extent was more limited than initially projected because NIFTP cannot be reliably diagnosed preoperatively, and institutions should establish their own category-specific ROM benchmarks given substantial practice variation.

Paper search

We performed a semantic search using the query "How did the 2017 WHO reclassification of noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) affect the risk of malignancy (ROM) estimates across Bethesda thyroid cytology categories in surgical cohorts?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 497 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Bethesda System Classification:** Does this study report thyroid cytology results using the Bethesda System for Reporting Thyroid Cytopathology?

- **Histopathological Correlation:** Does this study include surgical cohorts with histopathological correlation to determine malignancy rates?
- **NIFTP Impact Assessment:** Does this study specifically address or provide data that allows calculation of risk of malignancy (ROM) before and/or after NIFTP reclassification?
- **Publication Date:** Was this study published in 2009 or later?
- **Sample Size:** Does this study include a minimum of 50 cases with surgical correlation?
- **Study Design:** Is this study an observational study (cohort, cross-sectional, case series), systematic review, or meta-analysis?
- **Population Age:** Is this study focused on adult populations (age ≥ 18 years) or mixed populations that include adults?
- **Publication Type:** Is this a full-text peer-reviewed publication (not a conference abstract, editorial, or opinion piece)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design:**

Extract study design and setting details including:

- Study design (retrospective cohort, cross-sectional, etc.)
- Institution type (academic medical center, community hospital, etc.)
- Geographic location and population served
- Study timeframe (dates of data collection)
- Whether single-center or multi-center

- **Cohort Composition:**

Extract sample size and cohort characteristics including:

- Total number of thyroid FNA cases reviewed
- Number of cases with surgical resection/histologic follow-up
- Inclusion and exclusion criteria for the surgical cohort
- Patient demographics if provided (age, gender)
- Any selection bias considerations mentioned

- **NIFTP Cases:**

Extract details about NIFTP identification and distribution including:

- Total number of NIFTP cases identified
- Method used to identify/reclassify NIFTP cases
- Distribution of NIFTP cases across Bethesda categories (how many in each category II-VI)
- Percentage of all thyroid malignancies that were NIFTP
- Any challenges or criteria used for NIFTP diagnosis

- **ROM Before Reclassification:**

Extract risk of malignancy data with NIFTP counted as malignant including:

- ROM percentage for each Bethesda category (I through VI)
- Number of malignant cases and total cases per category
- Definition of malignancy used (what histologic diagnoses were included)
- Any confidence intervals or statistical measures provided

• **ROM After Reclassification:**

Extract risk of malignancy data with NIFTP counted as non-malignant including:

- ROM percentage for each Bethesda category (I through VI)
- Number of malignant cases (excluding NIFTP) and total cases per category
- Any confidence intervals or statistical measures provided
- How NIFTP was categorized in the revised analysis (uncertain, non-malignant, etc.)

• **ROM Changes:**

Extract the calculated changes in ROM including:

- Absolute ROM decrease for each Bethesda category (percentage points)
- Relative ROM decrease for each Bethesda category (percentage change)
- Which categories showed statistically significant changes (p-values)
- Categories most affected by the reclassification
- Overall impact on ROM across all categories

• **Clinical Implications:**

Extract authors' discussion of clinical impact including:

- Implications for surgical decision-making
- Effect on overtreatment concerns
- Recommendations for clinical practice
- Comparison to published Bethesda ROM benchmarks
- Any geographic or population-specific considerations
- Limitations of the reclassification noted

Characteristics of Included Studies

This systematic review encompasses 40 sources examining the impact of NIFTP reclassification on risk of malignancy across Bethesda categories. The included studies span multiple geographic regions and institutional settings, utilizing various methodological approaches.

Study	Full Text Retrieved?	Study Type	Geographic Location	Study Period	Sample Size (FNA/Resected)
H. Vuong et al., 2019	No	Meta-analysis	Multi-center	2016-2018	14,153 resected nodules
Melissa L. Mao et al., 2018	No	Retrospective cohort	Not specified	2010-2016	847 thyroidectomies

Study	Full Text Retrieved?	Study Type	Geographic Location	Study Period	Sample Size (FNA/Resected)
M. Bongiovanni et al., 2019	No	Systematic review/meta-analysis	Multi-center	Literature search Nov 2018	13,752 thyroidectomies
Simon Sung et al., 2019	No	Retrospective cohort	Not specified	2013-2017	4,500 FNA / 479 resected
Brenessa Lindeman et al., 2018	No	Retrospective cohort	Tertiary-care hospital	Apr 2016-Feb 2017	353 thyroidectomies
M. Bongiovanni et al., 2019a	Yes	Systematic review/meta-analysis	USA, Canada, Europe, Asia	Updated June 2018	915 NIFTP cases
E. Haaga et al., 2021	No	Systematic review/meta-analysis	Multi-center	Literature search 2020	25,892 surgical cases
Bryan Wei Wen Lee et al., 2021	No	Retrospective cohort	Singapore	2010-2016	821 FNA / 788 patients
R. Rosenblum et al., 2019	No	Retrospective cohort	Israel	2013-2015	3,701 nodules / 2,919 FNAs
I. Resta et al., 2023	No	Retrospective cohort	Not specified	2017-2022	247 FNA / 379 cases
R. Lastra et al., 2017	No	Retrospective cohort	Multi-center (8 institutions)	1-year period	2,226 surgeries / 119 nodules
Bayan A Alzumaili et al., 2023	No	Retrospective cohort	Not specified	2016-2022	255 FNA / 319 cases
P. Valderrabano et al., 2017	No	Retrospective cohort	Academic Cancer Center	2014-2016	190 ITNs / 102 resected
M. Pusztaszeri et al., 2017	No	Retrospective cohort	Not specified	2005-2015	86 NIFTP cases
P. Rosário et al., 2020	No	Prospective cohort	Not specified	Not specified	389 FNA / 323 resected
E. Chaigneau et al., 2018	No	Retrospective cohort	Not specified	2010-2016	602 cases
Ryan P. Lau et al., 2017	No	Retrospective cohort	Academic medical center	2013-2016	750 FNA with resection
Mara Ventura et al., 2019	No	Retrospective cohort	Not specified	2012-2014	5,625 FNA / 348 resected
Samantha M. Linhares et al., 2020	No	Retrospective cohort	Single-center	Not specified	565 FNA with thyroidectomy
Milim Kim et al., 2018	Yes	Retrospective cohort	South Korea	2012-2014	5,549 FNA / 1,891 resected
J. Kopczyński et al., 2020	No	Retrospective cohort	Not specified	Not specified	998 nodules

Study	Full Text Retrieved?	Study Type	Geographic Location	Study Period	Sample Size (FNA/Resected)
Darin Ruanpeng et al., 2019	No	Systematic review	Multi-center	Until Nov 2018	1,563 NIFTP cases
C. Kiernan et al., 2018	No	Retrospective cohort	Not specified	Not specified	1,046 FNA / 44 resected
H. Al-Maghrabi et al., 2022	No	Retrospective cohort	Saudi Arabia	2013-2017	1,066 FNA / 281 resected
K. Strickland et al., 2015	No	Retrospective cohort	Not specified	22-month period	655 FNA with resection
Prerna Guleria et al., 2019	No	Retrospective cohort	Tertiary care centre	4.5-year period	2,554 FNA / 81 resected
Shaham Beg et al., 2022	No	Retrospective cohort	Not specified	11-year period	244 resected
R. Mainthia et al., 2018	No	Retrospective cohort	Not specified	2006-2016	1,335 PTC / 164 NIFTP
Zhengfan Xu et al., 2024	No	Retrospective cohort	Not specified	2018-2022	53 FNA / 1,282 resected
M. Paja et al., 2021	No	Retrospective cohort	Iberian Peninsula	2005-2015	Multi-center (17 hospitals)
Hao Li et al., 2025	No	Retrospective chart review	Tertiary general hospital	2010-2016	243 nodules
Caroline Bourque et al., 2025	No	Retrospective analysis	Not specified	Not specified	114 AUS cases
Niti Sureka et al., 2025	No	Retrospective cohort	Tertiary care centre	2023-2024	387 FNA / 190 resected
J. Kopczyński et al., 2019	No	Retrospective cohort	Not specified	Not specified	998 nodules
F. Maletta et al., 2016	Yes	Retrospective cohort	Italy (multi-center)	Prior to 2016	96 NIFTP cases
Y. Nikiforov et al., 2016	Yes	Retrospective cohort	International (5 countries)	10-26 years follow-up	268 tumors (109 noninvasive)
J. Sauter et al., 2019	No	Retrospective cohort	Not specified	Median 4.4-9.9 years follow-up	1,277-1,616 FNAs
A. Bychkov et al., 2018	Yes	Retrospective cohort	5 Asian countries	1-3 years	11,372 FNA / 2,044 resected
Richa Singh et al., 2017	No	Retrospective cohort	Tertiary referral center	9-year period	177 FNA / 174 resected
H. Y. Na et al., 2019	No	Retrospective cohort	Not specified	Not specified	2,687 CNB / 946 resected

The majority of included studies were single-center retrospective cohort analyses, with five systematic reviews or meta-analyses providing pooled estimates . Geographic representation included studies from North America, Europe, Asia (including South Korea, Singapore, and multiple Asian countries), the Middle East (Israel, Saudi Arabia), and

the Iberian Peninsula . Sample sizes ranged from small single-institution series of approximately 100 cases to large meta-analyses pooling over 25,000 surgically resected cases .

NIFTP Prevalence and Distribution

Prevalence Among Surgical Cohorts

The prevalence of NIFTP among surgically resected thyroid nodules varied considerably across studies. The largest meta-analysis, including 25,892 surgical cases, found NIFTP constituted 4.4% (95% CI: 3.5–5.4%) of all resected cases . Individual institutional series reported prevalence ranging from 1.2% in Singapore to 8.5% in a more recent 2024 study . When expressed as a proportion of malignant diagnoses, NIFTP comprised 2.9% of all excised nodules and 5.3% of malignant nodules in the Asian multi-institutional study , 13% of all malignant thyroid neoplasms in one US series , and as high as 24.6% of malignancies in another cohort . Among papillary thyroid carcinomas specifically, NIFTP represented 1.5% of all PTCs in Korea , 5.2% of PTC cases in the Iberian Peninsula , 6.3% to 6.6% of thyroid surgeries in multi-institutional studies , and 13.8% of cases formerly diagnosed as PTC at one center .

Distribution Across Bethesda Categories

Three meta-analyses provided pooled estimates of NIFTP distribution across cytologic categories. The pooled distribution from 58 studies with 2,553 NIFTP cases showed: non-diagnostic 1.3% (95% CI: 0.8–1.7%), benign 8.9% (95% CI: 6.9–10.8%), AUS/FLUS 29.2% (95% CI: 25.0–33.4%), FN/SFN 24.2% (95% CI: 19.6–28.9%), suspicious for malignancy 19.5% (95% CI: 16.1–22.9%), and malignant 6.9% (95% CI: 5.2–8.7%) . A separate meta-analysis of 915 NIFTP cases found similar distributions: non-diagnostic 3%, benign 10%, AUS/FLUS 30%, FN/SFN 21%, suspicious for malignancy 24%, and malignant 8% . A third systematic review of 1,563 cases reported: non-diagnostic 3.6%, benign 10.0%, AUS/FLUS 34.2%, FN/SFN 22.7%, suspicious for malignancy 22.4%, and malignant 7.5% .

The majority of NIFTP cases (approximately 79%) fell within the intermediate cytologic categories (Bethesda III, IV, and V) . Individual studies consistently reported that indeterminate FNA results were most common for NIFTP, with 71.4% demonstrating indeterminate findings in one series and the majority classified as AUS/FLUS (42.9%) or FN/SFN (31.0%) in another multi-institutional study .

Effects on Risk of Malignancy by Bethesda Category

Meta-Analytic Findings

Meta-Analysis	Category Most Affected	Absolute ROM Decrease	Relative ROM Decrease	Overall ROM Change
H. Vuong et al., 2019	Category V (largest absolute)	16% (95% CI: 8-24)	32% (95% CI: 24-39) for Category III	Significant reduction in all categories except Category I
M. Bongiovanni et al., 2019	Category V	14% for Bethesda V; 3% for Bethesda VI	Not reported	5.5% pooled risk difference

Meta-Analysis	Category Most Affected	Absolute ROM Decrease	Relative ROM Decrease	Overall ROM Change
E. Haaga et al., 2021	AUS/FLUS and FN	ND: 2.4%; Benign: 2.7%; AUS/FLUS: 8.2%; FN: 8.2%; SM: 7.3%; Malignant: 1.1%	Not reported	Reduction across all categories

The meta-analysis by Vuong et al. found that the largest absolute decrease in ROM occurred in Bethesda category V (16%; 95% CI = 8 to 24), while the largest relative decrease was observed in category III (32%; 95% CI = 24 to 39) . Significant reductions were observed across all Bethesda categories except category I .

Single-Institution Studies: Bethesda Category III (AUS/FLUS)

Study	ROM Before	ROM After	Absolute Decrease	Statistical Significance
Lindeman et al., 2018	33%	18%	15 pp	p < 0.05
Sung et al., 2019	Not reported	Not reported	15%	p = 0.03
Lau et al., 2017	42.3%	22.3%	20 pp	Not reported
Linhares et al., 2020	55%	48%	7 pp	Not significant
Kim et al., 2018	75.8%	71.2%	4.6 pp	Not significant
Al-Maghrabi et al., 2022	50%	39.2%	10.8 pp	p = 0.2089
Bychkov et al., 2018	44.7%	38.6%	6.1 pp	Not significant
Ventura et al., 2019	Not specified	Not specified	5.5% absolute; 15.2% relative	Not reported
Guleria et al., 2019	51.8%	35.8%	16 pp (31% relative)	Not reported

Category III showed variable impact across studies. While some institutions demonstrated substantial decreases reaching 20 percentage points , others showed minimal changes of less than 5 percentage points . Statistically significant decreases were reported in two studies .

Single-Institution Studies: Bethesda Category IV (FN/SFN)

Study	ROM Before	ROM After	Absolute Decrease	Statistical Significance
Lindeman et al., 2018	29%	22%	7 pp	Not significant
Sung et al., 2019	Not reported	Not reported	16.2%	p = 0.04
Lau et al., 2017	48.7%	17.9%	30.8 pp	Not reported
Linhares et al., 2020	50%	35%	15 pp	Not significant

Study	ROM Before	ROM After	Absolute Decrease	Statistical Significance
Al-Maghrabi et al., 2022	53.8%	33.3%	20.5 pp	p = 0.0336
Bychkov et al., 2018	40.4%	30.6%	9.8 pp	p = 0.06
Ventura et al., 2019	Not specified	Not specified	1.8% absolute; 7.6% relative	Not reported

Category IV demonstrated the most consistent substantial decreases, with multiple studies showing absolute reductions exceeding 15 percentage points . The FN/SFN category was identified as the only category showing greater than 20% relative reduction in the Asian multi-center study . Statistically significant decreases were achieved in two studies .

Single-Institution Studies: Bethesda Category V (Suspicious for Malignancy)

Study	ROM Before	ROM After	Absolute Decrease	Statistical Significance
Lindeman et al., 2018	91%	73%	18 pp	p < 0.05
Lau et al., 2017	93.6%	61.7%	31.9 pp	Not reported
Linhares et al., 2020	93%	91%	2 pp	Not significant
Bychkov et al., 2018	88.0%	83.1%	4.9 pp	Not significant
Kiernan et al., 2018	68%	60%	8 pp	Not reported
Ventura et al., 2019	Not specified	Not specified	10.3% absolute; 14.2% relative	Not reported
Strickland et al., 2015	87.2%	45.7%	41.5 pp (48% relative)	Not reported

Category V exhibited the most dramatic range of ROM changes, with decreases ranging from 2 percentage points to nearly 42 percentage points . The Strickland et al. study, which evaluated the impact if noninvasive follicular variant of papillary thyroid carcinoma were no longer termed carcinoma, found the most clinically significant decrease in this category with a relative reduction of nearly 50% .

Single-Institution Studies: Bethesda Categories I, II, and VI

Changes in categories I (non-diagnostic), II (benign), and VI (malignant) were generally modest. For category II (benign), ROM decreased from 5.5% to 2.5% , from 18% to 14% , from 21.1% to 11.9% (p = 0.0343) , and from 13.2% to 5.4% across different studies. Category VI showed minimal changes, decreasing from 100% to 97% , from 99% to 98% , and from 95.5% to 88% . Category I often showed no change or minimal decreases .

Overall Impact on Institutional ROM

The overall decrease in malignancy rates across entire cohorts ranged from 2% (31% to 29%) to 7.3% . For the subgroup with indeterminate cytology (Bethesda III-V), the decrease was more pronounced at 13.1% . The overall ROM across

all categories decreased from 58% to 55% in one series and changed marginally despite statistically significant changes in individual categories in others .

Studies Using Revised 2018 NIFTP Criteria

One study specifically evaluated ROM using the revised 2018 diagnostic criteria for NIFTP classification, which were proposed to avoid misdiagnosis of papillary thyroid cancer as NIFTP . This study found only 5 cases (2.3% of all papillary thyroid carcinoma diagnoses) met the revised criteria , with ROM decreases of 0.7% in FN/SFN, 4.3% in SM, and 0.5% in malignant categories—none of which were statistically significant .

Synthesis

Explaining Heterogeneity in ROM Changes

The substantial variability in ROM changes across studies (ranging from negligible to nearly 50% relative decreases in some categories) can be attributed to several key factors.

Geographic and Population Differences

A consistent finding across multiple studies was that NIFTP prevalence and its impact on ROM differed markedly between Western and Asian populations. The meta-analysis by Vuong et al. found that decreases in ROM were more prominent in Western than in Asian cohorts . The multi-institutional Asian study (Thailand, India, Taiwan, Japan, Korea) found NIFTP constituted only 2.9% of excised nodules and 5.3% of malignant nodules , with the magnitude of impact "much lower than in the Western reports" . Similarly, the Singapore study found that "the impact of NIFTP on ROM in our institution also appears to be lower than that reported in the Western studies" , with NIFTP incidence of only 1.2% . This geographic variation appears related to differences in the baseline prevalence of follicular variant papillary thyroid carcinoma in different populations.

Baseline ROM and Surgical Selection Practices

Studies with higher baseline ROM values tended to show larger absolute decreases. For example, institutions with high surgical resection rates for indeterminate nodules demonstrated higher NIFTP rates . There was a positive correlation between the rate of NIFTP and resection rate ($r = 0.83$; $P = 0.02$) . Institutions that operate more frequently on indeterminate nodules capture more NIFTP cases, amplifying the apparent impact of reclassification on ROM.

Diagnostic Criteria and Pathologist Interpretation

The Iberian Peninsula multi-center study specifically identified pathologist interpretation of nuclear alterations as the main cause of inter-institutional variability in NIFTP rates (range: 0-12.1%) . The NIFTP rate showed linear correlation with total PTC ($p = 0.03$) and FVPTC ($p = 0.007$) identified at each center . The application of stricter 2018 diagnostic criteria for NIFTP substantially reduced the number of qualifying cases, with only 2.3% of papillary thyroid carcinomas meeting revised criteria compared to the original estimates .

Study Period and Diagnostic Evolution

Earlier studies (pre-2017) that retrospectively reclassified encapsulated follicular variant PTCs as NIFTP showed higher NIFTP prevalence rates (14.5% to 24.6% of malignancies) , while prospective studies applying strict criteria from the outset found lower rates . The landmark Nikiforov et al. 2016 study that introduced NIFTP terminology found NIFTP represented 13.8% of cases formerly diagnosed as PTC at one center and estimated the reclassification would affect more than 45,000 patients worldwide each year .

Context-Specific Conclusions

Western Institutions with High Resection Rates : Expect substantial ROM decreases, particularly in Bethesda categories III (15-20 percentage points) , IV (15-30 percentage points) , and V (18-40 percentage points) . These institutions should counsel patients toward more conservative surgical options when NIFTP is in the differential .

Asian Institutions : May not experience significant effects on thyroid cytopathology practice . ROM decreases are generally modest (less than 10 percentage points) and often statistically non-significant . The NIFTP entity has very little impact on ROM for diagnostic categories in these settings .

Institutions Using Strict 2018 Criteria : Experience minimal ROM changes across all categories , as the revised criteria substantially reduce the number of cases qualifying as NIFTP.

Clinical Practice Implications

Multiple studies confirmed that NIFTP is a low-risk neoplasm with indolent clinical behavior . Follow-up data demonstrated no structural persistence, recurrences, or metastases attributable to isolated NIFTP . Patients with NIFTP who underwent lobectomy required no completion thyroidectomy and received no postoperative radioactive iodine, with no recurrence or metastases recorded at median follow-up of 35 months . The original Nikiforov et al. study reported that all 109 patients with noninvasive EFVPTC (67 treated with lobectomy alone, none received radioactive iodine ablation) were alive with no evidence of disease after median follow-up of 13 years .

The reclassification was intended to reduce overtreatment , though the actual surgical impact was more limited than initially projected. One study found that while 14.5% of PTCs met NIFTP criteria, the impact with regard to extent of surgery was limited to 3.2% of patients compared to the projected potential impact of 18.6% . This is because NIFTP cannot be reliably diagnosed preoperatively—cytologic features overlap with invasive follicular variant PTC , and NIFTP shows no characteristic features on ultrasound . The majority of NIFTP will be classified as indeterminate lesions by cytology or molecular testing, meaning thyroidectomy remains a common treatment modality .

Surgeons should assess their respective institution's experiences with NIFTP and the Bethesda System , and treatment centers should define their own ROM for each category given the substantial inter-institutional variability .

References

- A. Bychkov, S. Keelawat, S. Agarwal, D. Jain, C. Jung, Soonwon Hong, Chiung-Ru Lai, S. Satoh, and K. Kakudo. “Impact of Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features on the Bethesda System for Reporting Thyroid Cytopathology: A Multi-Institutional Study in Five Asian Countries.” *Pathology (Sydney)*, 2018.
- Bayan A Alzumaili, Lauren N. Krumeich, Reagan D. Collins, T. Kravchenko, Emad Ababneh, Adam S. Fisch, W. Faquin, et al. “A Comprehensive Study on the Diagnosis and Management of Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP).” *Thyroid*, 2023.
- Brenessa Lindeman, Matthew A. Nehs, Trevor E. Angell, E. Alexander, A. Gawande, F. Moore, G. Doherty, and N. Cho. “Effect of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP) on Malignancy Rates in Thyroid Nodules: How to Counsel Patients on Extent of Surgery.” *Annals of Surgical Oncology*, 2018.
- Bryan Wei Wen Lee, M. Bundeale, R. Tan, E. Fu, A. Chew, J. Wong, C. H. C. Siew, et al. “Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features and the Risk of Malignancy in Thyroid Cytology: Data from Singapore.” *Annals of the Academy of Medicine, Singapore*, 2021.

- C. Kiernan, V. Weiss, M. Mehrad, K. Ely, Naira Baregamian, and C. Solórzano. "New Terminology—Noninvasive Follicular Neoplasm with Papillary-like Nuclear Features (NIFTP) and Its Effect on the Rate of Malignancy at a Single Institution." *Surgery*, 2018.
- Caroline Bourque, Gianluca Savoia, Maxine Noik, Livia Florianova, Saruchi Bandargal, S. D. da Silva, Rick Payne, and M. Pusztaszeri. "Malignancy Risk, Molecular Mutations, and Surgical Outcomes of Thyroid Nodules Classified as Atypia of Undetermined Significance in the Bethesda System: A Comprehensive Analysis." *Cytopathology*, 2025.
- Darin Ruanpeng, W. Cheungpasitporn, C. Thongprayoon, J. Hennessey, and Rupendra T. Shrestha. "Systematic Review and Meta-Analysis of the Impact of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP) on Cytological Diagnosis and Thyroid Cancer Prevalence." *Endocrine Pathology*, 2019.
- E. Chaigneau, G. Russ, B. Royer, C. Bigorgne, M. Bienvenu-Perrard, A. Rouxel, L. Leenhardt, L. Belin, and C. Buffet. "TIRADS Score Is of Limited Clinical Value for Risk Stratification of Indeterminate Cytological Results." *European Journal of Endocrinology*, 2018.
- E. Haaga, D. Kalfeřt, M. Ludvikova, and I. Kholová. "Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features Is Not a Cytological Diagnosis, but It Influences Cytological Diagnosis Outcomes: A Systematic Review and Meta-Analysis." *Acta Cytologica*, 2021.
- F. Maletta, Federica Massa, L. Torregrossa, Eleonora Duregon, G. Casadei, F. Basolo, G. Tallini, M. Volante, Y. Niki-forov, and M. Papotti. "Cytological Features of "Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features" and Their Correlation with Tumor Histology." *Human Pathology*, 2016.
- H. Al-Maghrabi, Mohamed Tashkandi, Waleed Khayyat, A. Alghamdi, Mohammed Alsalmi, Alhussain J. Alzahrani, H. Al-Hakami, and Mohammed Alqarni. "Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP) Lowers the Risk of Malignancy in the Bethesda System for Reporting Thyroid Cytopathology Diagnostic Categories." *Saudi Journal of Medicine and Medical Sciences*, 2022.
- H. Vuong, T. T. K. Tran, A. Bychkov, C. Jung, T. Nakazawa, K. Kakudo, R. Katoh, and T. Kondo. "CLINICAL IMPACT OF NON-INVASIVE FOLLICULAR THYROID NEOPLASM WITH PAPILLARY-LIKE NUCLEAR FEATURES ON THE RISK OF MALIGNANCY IN THE BETHESDA SYSTEM FOR REPORTING THYROID CYTOPATHOLOGY: A META-ANALYSIS OF 14,153 RESECTED THYROID NODULES." *Endocrine Practice*, 2019.
- H. Y. Na, J. Woo, J. Moon, J. Choi, Woo-Jin Jeong, Yeo-Koon Kim, J. Choe, and S. Y. Park. "Preoperative Diagnostic Categories of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features in Thyroid Core Needle Biopsy and Its Impact on Risk of Malignancy." *Endocrine Pathology*, 2019.
- Hao Li, Caroline C Y Lee, H. W. Khoo, M. Bunde, Jerome Y J Gan, Ernest W Z Fu, M. Y. Lim, J. Goh, and Alvin Y Q Soon. "Refining the Management of Bethesda III Thyroid Nodules by Combining Nuclear Atypia with Sonographic Suspicion." *Otolaryngology Head & Neck Surgery*, 2025.
- I. Resta, M. Gubbiotti, K. Montone, V. Livolsi, and Z. Baloch. "An Investigation into Non-Invasive Follicular Thyroid Neoplasms with Papillary-Like Nuclear Features (NIFTP): Does the Initial Proposal on NIFTP Behavior Hold True?" *Human Pathology*, 2023.
- J. Kopczyński, Agnieszka Suligowska, K. Niemyska, I. Pałyga, A. Walczyk, Danuta Gąsior-Perczak, A. Kowalik, et al. "Did Introducing a New Category of Thyroid Tumors (Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features) Decrease the Risk of Malignancy for the Diagnostic Categories in the Bethesda System for Reporting Thyroid Cytopathology?" *Endocrine Pathology*, 2020.
- , et al. "Influence of the Reclassification of NIFTP as a Uncertain Tumor on Risk of Malignancy for the Diagnostic Categories According to the Bethesda System for Reporting Thyroid Cytopathology." *Endokrynologia Polska*, 2019.
- J. Sauter, Heidi D. Lehrke, Xiaotun Zhang, Osamah T Al Badri, R. Rodríguez-Gutiérrez, D. Delivanis, Naykky M Singh Ospina, et al. "Assessment of The Bethesda System for Reporting Thyroid Cytopathology." *American Journal of Clinical Pathology*, 2019.
- K. Strickland, B. Howitt, E. Marqusee, E. Alexander, E. Cibas, J. Krane, and J. Barletta. "The Impact of Noninvasive

- Follicular Variant of Papillary Thyroid Carcinoma on Rates of Malignancy for Fine-Needle Aspiration Diagnostic Categories.” *Thyroid*, 2015.
- M. Bongiovanni, L. Giovanella, F. Romanelli, and P. Trimboli. “Cytological Diagnoses Associated with Noninvasive Follicular Thyroid Neoplasms with Papillary-Like Nuclear Features According to the Bethesda System for Reporting Thyroid Cytopathology: A Systematic Review and Meta-Analysis.” *Thyroid*, 2019.
- M. Bongiovanni, Wc Faquin, L. Giovanella, C. Durante, P. Kopp, and P. Trimboli. “Impact of Non-Invasive Follicular Thyroid Neoplasms with Papillary-Like Nuclear Features (NIFTP) on Risk of Malignancy in Patients Undergoing Lobectomy/Thyroidectomy for Suspicious for Malignancy or Malignant Fine-Needle Aspiration Cytology Findings: A Systematic Review and Meta-Analysis.” *European Journal of Endocrinology*, 2019.
- M. Paja, C. Zafón, C. Iglesias, A. Ugalde, J. Cameselle-Teijeiro, Gemma Rodríguez-Carnero, P. Fernández-Seara, et al. “Rate of Non-Invasive Follicular Thyroid Neoplasms with Papillary-Like Nuclear Features Depends on Pathologist’s Criteria: A Multicentre Retrospective Southern European Study with Prolonged Follow-up.” *Endocrine*, 2021.
- M. Pusztaszeri, F. Triponez, P. Meyer, and S. Sadowski. “Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP): Report of an Institutional Experience with 86 Cases,” 2017.
- Mara Ventura, M. Melo, G. Fernandes, and F. Carrilho. “RISK OF MALIGNANCY IN THYROID CYTOLOGY: THE IMPACT OF THE RECLASSIFICATION OF NONINVASIVE FOLLICULAR THYROID NEOPLASM WITH PAPILLARY-LIKE NUCLEAR FEATURES (NIFTP).” *Endocrine Practice*, 2019.
- Melissa L. Mao, Thomas Joyal, Omar Picado, Darcy A. Kerr, J. Lew, and Josefina C. Farrá. “Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features Reclassification and Its Impact on Thyroid Malignancy Rate and Treatment.” *Journal of Surgical Research*, 2018.
- Milim Kim, Joung Eun Kim, Hyun Jeong Kim, Yul Ri Chung, Y. Kwak, and S. Y. Park. “Cytologic Diagnosis of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features and Its Impact on the Risk of Malignancy in the Bethesda System for Reporting Thyroid Cytopathology: An Institutional Experience.” *Journal of Pathology and Translational Medicine*, 2018.
- Niti Sureka, C. Ahluwalia, S. Ahuja, NehaKawatra Madan, Meetu Agrawal, and Sunil Ranga. “Assessment of Thyroid Fine-Needle Aspirates Using 2023 Bethesda System.” *Acta Cytologica*, 2025.
- P. Rosário, Alexandre Lemos Silva, and M. Calsolari. “The ATA Classification and TI-RADS ACR Predict Not Only Benignity but Also the Histology of Nonbenign Tumors in Thyroid Nodules with Indeterminate Cytology.” *Diagnostic Cytopathology*, 2020.
- P. Valderrabano, Laila Khazai, Marino E. Leon, Z. Thompson, Zhenjun Ma, C. Chung, J. Hallanger-Johnson, et al. “Evaluation of ThyroSeq V2 Performance in Thyroid Nodules with Indeterminate Cytology.” *Endocrine-Related Cancer*, 2017.
- Perna Guleria, S. Agarwal, V. Iyer, D. Jain, S. Mathur, and D. Yadav. “Subcategorisation of AUS/FLUS Thyroid Lesions as Per the 2017 Bethesda System for Reporting Thyroid Cytopathology: A Retrospective Study from a Tertiary Care Centre Analysing Risk of Malignancy (ROM) of the Different Subcategories.” *Journal of Clinical Pathology*, 2019.
- R. Lastra, G. Birdsong, D. Hwang, M. Jordá, Darcy A. Kerr, C. McGrath, Shelley I. Odronic, et al. “Preoperative Cytologic Interpretation of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features: A 1-Year Multi-Institutional Experience.” *Journal of the American Society of Cytopathology*, 2017.
- R. Mainthia, H. Wachtel, Yufei Chen, E. Mort, S. Parangi, P. Sadow, and C. Lubitz. “Evaluating the Projected Surgical Impact of Reclassifying Noninvasive Encapsulated Follicular Variant of Papillary Thyroid Cancer as Noninvasive Follicular Thyroid Neoplasm with Papillary-like Nuclear Features.” *Surgery*, 2018.
- R. Rosenblum, A. Shtabsky, S. Marmor, L. Trejo, I. Yaish, S. Barnes, M. Yehuda, N. Stern, Z. Silman, and K. Tordjman. “Indeterminate Nodules by the Bethesda System for Reporting Thyroid Cytopathology in Israel: Frequency, and Risk of Malignancy After Reclassification of Follicular Thyroid Neoplasm with Papillary-Like Features.” *European*

- Journal of Surgical Oncology*, 2019.
- Richa Singh, Jorge Avila, K. Jo, Kevin T. K. Nguyen, N. Carrillo, E. Huang, and M. Campbell. "Patients with Non-Invasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features Are Unlikely to Have Malignant Pre-operative Cytology." *Annals of Surgical Oncology*, 2017.
- Ryan P. Lau, J. Paulsen, T. Brandler, Cheng Liu, A. Simsir, and Fang Zhou. "Impact of the Reclassification of "Non-invasive Encapsulated Follicular Variant of Papillary Thyroid Carcinoma" to "Noninvasive Follicular Thyroid Neoplasm With Papillary-Like Nuclear Features" on the Bethesda System for Reporting Thyroid Cytopathology: A Large Academic Institution's Experience." *American Journal of Clinical Pathology*, 2017.
- Samantha M. Linhares, Bryan W Whitfield, A. F. Lee, Daniela Gordillo, Omar Picado, Mohammed Jeraq, Josefina C. Farrá, and J. Lew. "Impact of Noninvasive Follicular Thyroid Neoplasm With Papillary-Like Nuclear Features on Revised Bethesda System Malignancy Rates at a Single Institution." *Journal of Surgical Research*, 2020.
- Shaham Beg, S. Khan, Isabelle H. Cui, T. Scognamiglio, and R. Rao. "Analysis of a Pre-2017 Follicular Variant Papillary Thyroid Carcinoma Cohort Reclassified as Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Features (NIFTP): A 11-Year Retrospective Single Institution Experience." *Journal of the American Society of Cytopathology*, 2022.
- Simon Sung, Elizabeth M Margolskee, D. Chen, and P. Tiscornia-Wasserman. "Incidence of Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features and Change in Risk of Malignancy for "The Bethesda System for Reporting Thyroid Cytology"." *Journal of the American Society of Cytopathology*, 2019.
- Y. Nikiforov, R. Seethala, G. Tallini, Z. Baloch, F. Basolo, L. Thompson, J. Barletta, et al. "Nomenclature Revision for Encapsulated Follicular Variant of Papillary Thyroid Carcinoma: A Paradigm Shift to Reduce Overtreatment of Indolent Tumors." *JAMA Oncology*, 2016.
- Zhengfan Xu, Alyssa Vitale, Christian Keller, Wamidh Alkhoory, Ziyang Zhang, and Lisi Yuan. "Noninvasive Follicular Thyroid Neoplasm with Papillary-Like Nuclear Features (NIFTP): Prevalence, Cyto-Histo Correlation, and Molecular and Ultrasonographic Profile." *Annals of Diagnostic Pathology*, 2024.